FILE 6: analytical_framework.

Description: Full specification of the statistical models, hypothesis testing framework, and model diagnostics used in the study.

Hypotheses:

- **H_1:** There is a statistically significant difference in the perceived significance of barriers across the eight adoption barrier categories.
  - Statistical form: $\bar{B}_1 \neq \bar{B}_2 \neq ... \neq \bar{B}_8$
  - Test: One-way ANOVA comparing mean barrier scores.
- **H_2:** There is a statistically significant association between stakeholder characteristics (professional category, experience, location) and perceptions of blockchain-modular construction barriers.
  - Statistical form: $\chi^2 >$ critical value
  - Test: Chi-square (χ^2) test of independence.
- **H_3:** Blockchain integration and modular construction adoption have a statistically significant positive effect on perceived housing affordability improvement.
  - Statistical form: $\beta_1 > 0$ and $\beta_2 > 0$
  - Test: Multiple linear regression.

Exploratory Factor Analysis (RQ1):

Principal Component Analysis with Varimax rotation applied to 30 barrier variables. Kaiser-Meyer-Olkin (KMO) measure = 0.87; Bartlett's test of sphericity $\chi^2 = 4521.3$, $p < 0.001$. Eight factors extracted with eigenvalues > 1.0, explaining 68.4% of total variance.

Chi-Square Test (RQ1):

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Where:

- χ^2 = Chi-square test statistic
- O_ij = observed frequency in cell (i,j)
- E_ij = expected frequency in cell (i,j)
- r = number of rows; c = number of columns

Multiple Linear Regression (RQ2):

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 \times X_2) + \beta_4 X_3 + \beta_5 X_4 + \epsilon$$

Where:

- Y = perceived housing affordability improvement
- X_1 = modular construction adoption score

- X_2 = blockchain integration score
- $X_1 \times X_2$ = interaction term
- X_3 = stakeholder category (dummy)
- X_4 = study location (dummy)
- β_0 = intercept; β_1–β_5 = regression coefficients; ε = error term

Model Results:

- $R^2 = 0.43$; Adjusted $R^2 = 0.42$
- $F(5, 314) = 28.47$, $p < 0.001$
- Modular adoption: $\beta = 0.31$, $t = 5.82$, $p < 0.001$
- Blockchain integration: $\beta = 0.28$, $t = 5.24$, $p < 0.001$
- Interaction: $\beta = 0.19$, $t = 3.15$, $p = 0.002$

Model Diagnostics:

- Multicollinearity: VIF < 2.0 (all predictors), threshold 5.0.
- Autocorrelation: Durbin-Watson = 1.92 (acceptable 1.5–2.5).
- Normality of residuals: Shapiro-Wilk W = 0.98, p = 0.12; Q-Q plot linear.
- Homoscedasticity: residual vs. predicted plot shows no pattern.

Software:

- SPSS Version 27 for all quantitative analyses.
- NVivo for qualitative thematic analysis.
- Microsoft Excel for descriptive statistics and data cleaning.